

Name: Jojo Greear Class: HS Engineering

MyDesign Portfolio

Element G Initial Template

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

Prototype Construction of Design #1

	Details	Tools and Manufacturing Methods used
Step 1	At first we took a piece of thick plywood and measured it and marked it so it was a little bigger than the platform. Then we took some thicker pieces of actual wood and cut them into posts.	Pieces of wood, yard stick, pencil, handsaw
Step 2	We drilled the posts into the platform then we did the same to the platform but that didn't work out so we hot glued it to the wood. And	Drill, screws, rolling pins, hot glue

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	drilled holes into the posts so we could put the rolling pins in on either side then hot glued them so they wouldn't fall put.	
Step 3	We drilled holes into the posts so we could put the rolling pins in on either side then hot glued them so they wouldn't fall out.	Drill, Drill bit, rolling pins,
Step 4	We got some black fabric and cut it so it was thin enough to fit on the rolling pins then we attached it to the rolling pins with staples but that didn't work so we taped it on there. We also put some screws on the top pieces of wood so it wouldn't move out of place.	Black fabric, scissors, screws, drill
Step 5	We put some gears onto the metal part of the rolling pin and then another onto the motors so it could get traction and move the rolling pins.	Gears, motors, hot glue
Step 6	We drilled two holes above the motors	Drill, drill bit, motor, zip ties

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	and attached the motors to the posts with zip ties.	
Step 7	The last step was to tidy up the wires and make sure they weren't going everywhere so we did that by taping it to places with electrical tape.	Black electrical tape

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
Plant tray	A plant tray that is 20x20 to hold our plant	donation	free	1	free
wood	We needed wood to build the frame	donation	free	8	free
Roller	We used the rollers to make the fabric move to cover and uncover the plant	donation	free	2	free
gears	We used gears to make the rollers move	donation	free	4	free

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	when the motor ran				
Batteries	We used this to power the motor	walmart	6.50	1	6.50
Motors	We used this to make the device function	amazon	15	2	30
prograner	We used this to program the about of time the device runs	walmart	12	1	12
screws	We used this to connect the wood together	Donation	free	9	free
fabric	We used this to cover the plant at the given time	donation	free	1	free
Zip ties	We used this to hold the motors in place	donation	free	2	free

Construction of the Final Prototype (G2)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
20x20 plant tray	It will help us know what the size of the frame needs to be to hold the tray.

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Solar or battery powered	It will help us find ways to power the device for a long amount of time.
Device functionality	This is to make sure it works and covers the plant when it needs to be covered.
Works in different environment	This is to make sure it can work strongly and effectively in each kind of environment.
Works specifically to each plant	This is to make sure the plant you are trying to protect gets the correct amount of sunlight
How it looks	This was to make sure people want to buy it and get peoples attention to the device
Cost of device	This was to make it affordable to people who want to purchase the device for their plants.

Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Attribute	Description	Describe your prototype's functionality to test this attribute
A rolling shade	A shade that rolls over top and off of the plant.	You can test it by pressing a button and the shade will either roll over or off of the plant
Easy to move	It's easy to move, it's decently small.	You can pick it up and move it and see it's easy to move.
Easy to use	Right now all you have to do is press a button but once it's programmed correctly you won't have to do anything.	Right now since it's a button you just press a button to see that it works, and that's how you could test it as well.



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Addressing Non-Testable Attributes (G4)

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Moves on its own	The project does not move on its own because it does not have wheels or anything to move it.
Shades all sides of the plant	The shade only shades the top part of the plant which should be enough